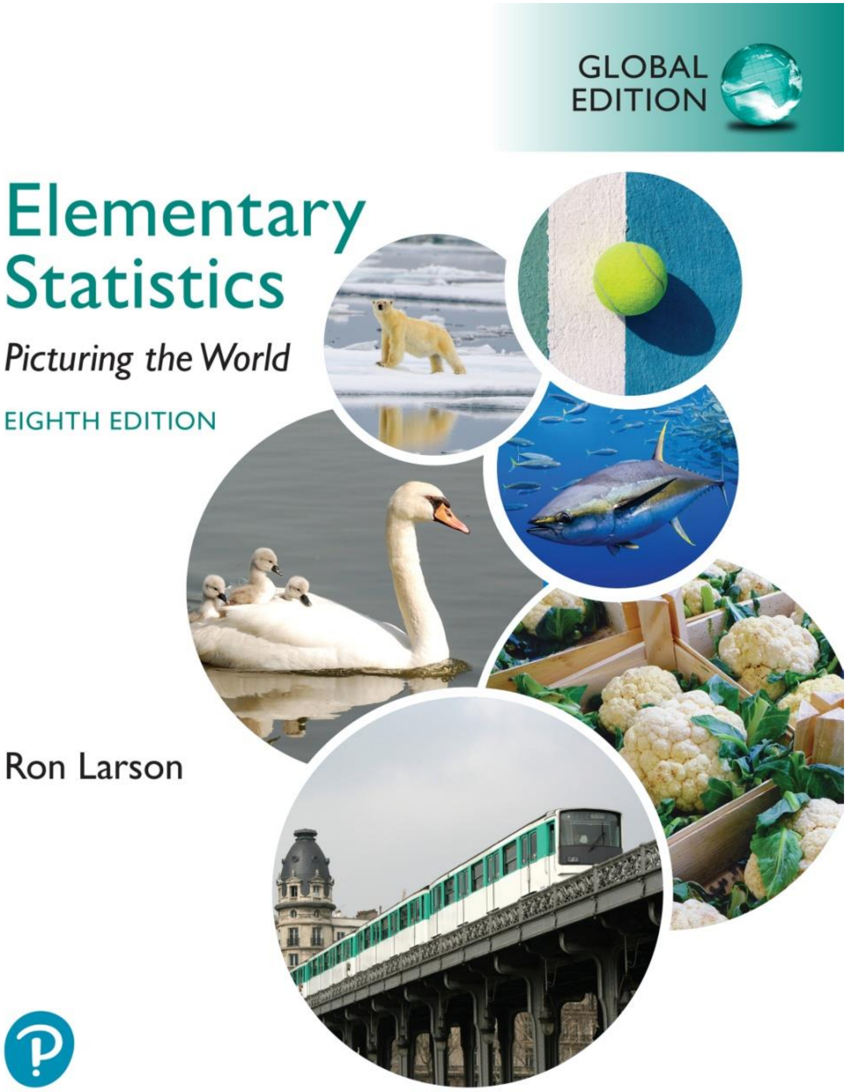
Chapter 5

Normal Probability Distributions



Chapter Outline

- 5.1 Introduction to Normal Distributions and the Standard Normal Distribution
- 5.2 Normal Distributions: Finding Probabilities
- 5.3 Normal Distributions: Finding Values
- 5.4 Sampling Distributions and the Central Limit Theorem
- 5.5 Normal Approximations to Binomial Distributions

Section 5.4

Sampling Distributions and the Central Limit Theorem

Section 5.4 Objectives

- How to find sampling distributions and verify their properties
- How to interpret the Central Limit Theorem
- How to apply the Central Limit Theorem to find the probability of a sample mean

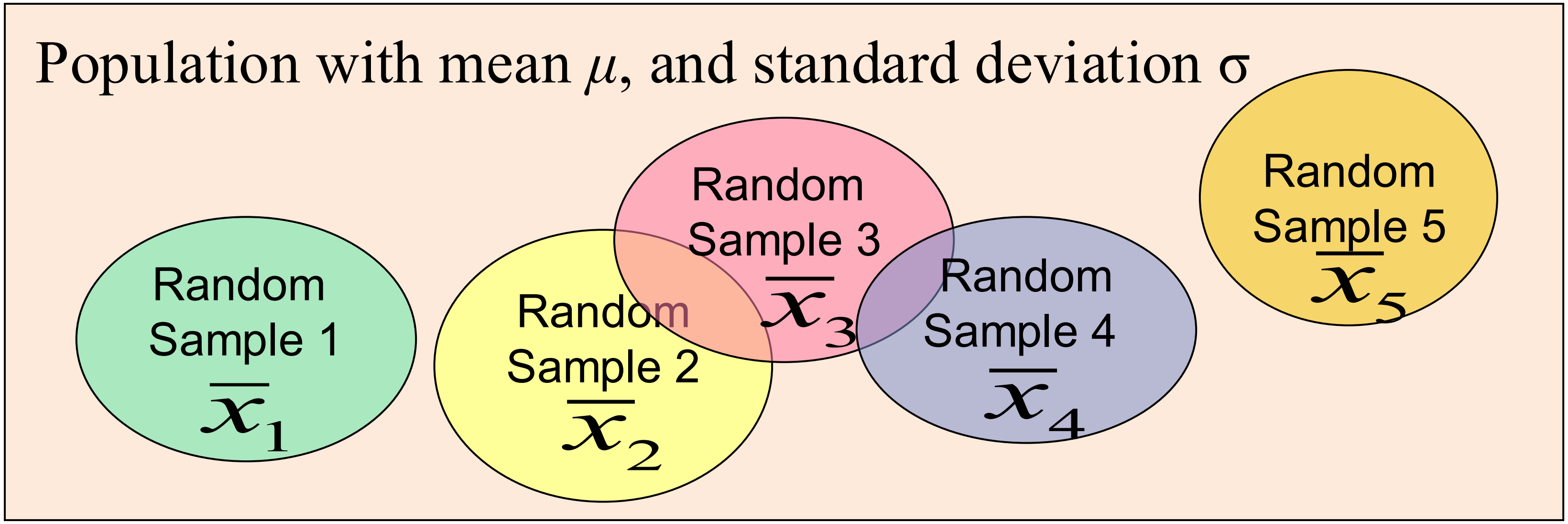
Sampling Distributions

Sampling distribution

- The probability distribution of a sample statistic that is formed when random samples of size *n* are repeatedly taken from a population.

- If the sample statistic is the sample mean, then the distribution is the **Sampling distribution of sample means.**

Sampling Distribution of Sample Means



The sampling distribution consists of the values of the sample means, $\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4, \bar{x}_5, \ldots$

Properties of Sampling Distributions of Sample Means

1. The mean of the sample means, $\mu_{\bar{x}}$, is equal to the population mean μ.

$$\mu_{\bar{x}} = \mu$$

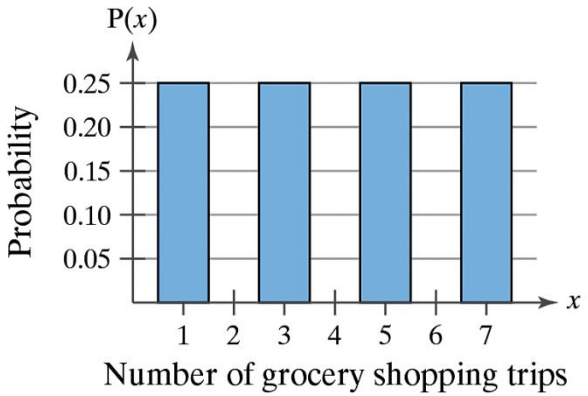
2. The standard deviation of the sample means, $\sigma_{\bar{x}}$, is equal to the population standard deviation, σ divided by the square root of the sample size, *n*.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

- Called the **standard error of the mean**.

Example: A Sampling Distribution of Sample Means

The number of times four people go grocery shopping in a month is given by the population values {1, 3, 5, 7}. A probability histogram for the data is shown. You randomly choose two of the four people, with replacement. List all possible samples of size $n = 2$ and calculate the mean of each. These means form the sampling distribution of the sample means. Find the mean, variance, and standard deviation of the sample means. Compare your results with the mean $\mu = 4$, variance $\sigma^2 = 5$, and standard deviation $\sigma = \sqrt{5} \approx 2.2$ of the population.



Solution: A Sampling Distribution of Sample Means

Solution:

List all 16 samples of size 2 from the population and the mean of each sample.

Sample	Sample mean, $\bar{x}$
1, 1	1
1, 3	2
1, 5	3
1, 7	4
3, 1	2
3, 3	3
3, 5	4
3, 7	5

Sample	Sample mean, $\bar{x}$
5, 1	3
5, 3	4
5, 5	5
5, 7	6
7, 1	4
7, 3	5
7, 5	6
7, 7	7

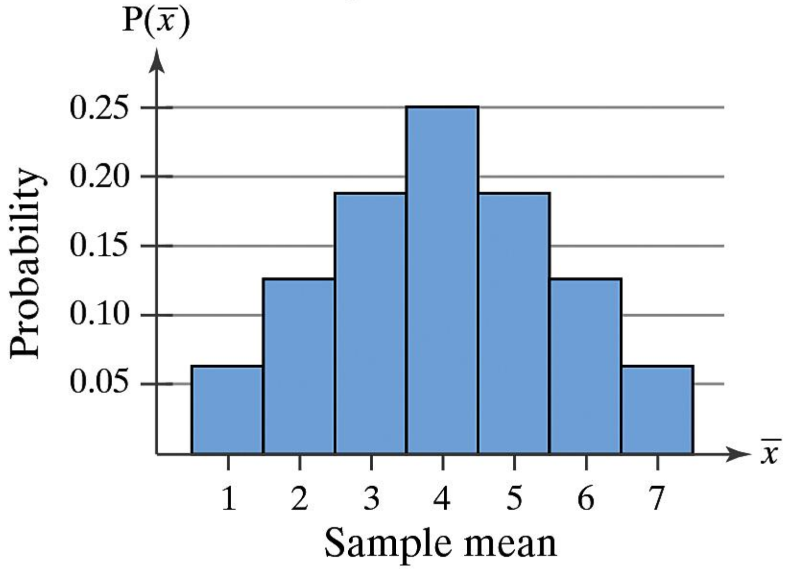
Solution: A Sampling Distribution of Sample Means

Construct a probability distribution of the sample means. Then, you can graph the sampling distribution using a probability histogram.

Probability Distribution of Sample Means

$\bar{x}$	f	Probability
1	1	$1/16$
2	2	$2/16$
3	3	$3/16$
4	4	$4/16$
5	3	$3/16$
6	2	$2/16$
7	1	$1/16$

Probability Histogram of Sampling Distribution of $\bar{x}$



Solution: A Sampling Distribution of Sample Means

The mean, variance, and standard deviation of the 16 sample means are

$\mu_{\bar{x}} = 4$ Mean of the sample means

$(\sigma_{\bar{x}})^2 = \dfrac{5}{2} = 2.5$ Variance of the sample means

and

$\sigma_{\bar{x}} = \sqrt{\dfrac{5}{2}} = \sqrt{2.5} \approx 1.6$ Standard deviation of the sample means

Solution: A Sampling Distribution of Sample Means

These results satisfy the properties of sampling distributions because

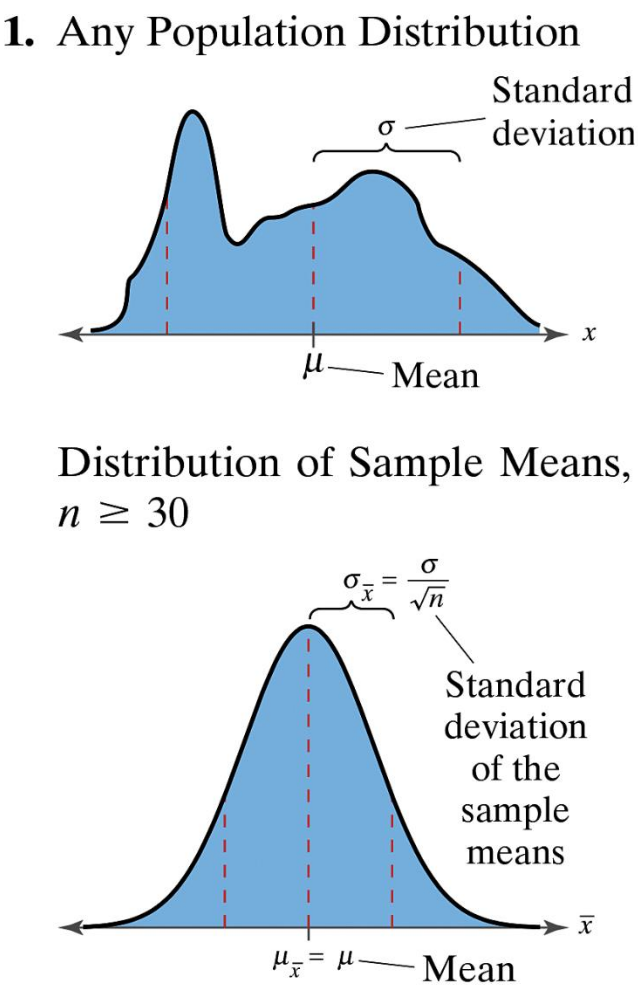
$$\mu_{\bar{x}} = \mu = 4$$

and

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{\sqrt{5}}{\sqrt{2}} \approx 1.6.$$

The Central Limit Theorem

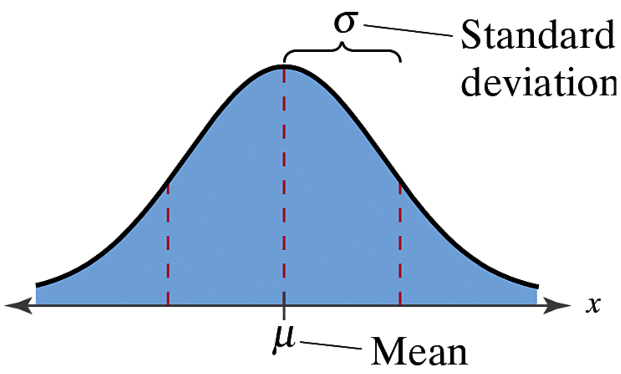
1. If samples of size $n \geq 30$, are drawn from any population with mean = μ and standard deviation = σ, then the sampling distribution of the sample means approximates a normal distribution. The greater the sample size, the better the approximation.



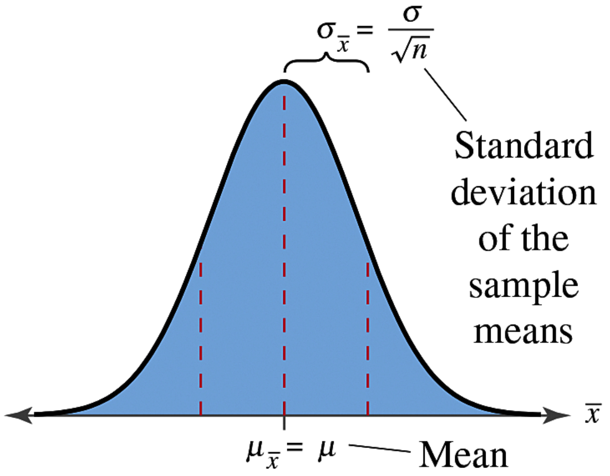
The Central Limit Theorem

2. If the population itself is normally distributed, the sampling distribution of the sample means is normally distribution for ***any*** sample size n.

2. Normal Population Distribution



Distribution of Sample Means (any n)



The Central Limit Theorem

- In either case, the sampling distribution of sample means has a mean equal to the population mean.

$$\mu_{\bar{x}} = \mu \qquad \text{Mean of the sample means}$$

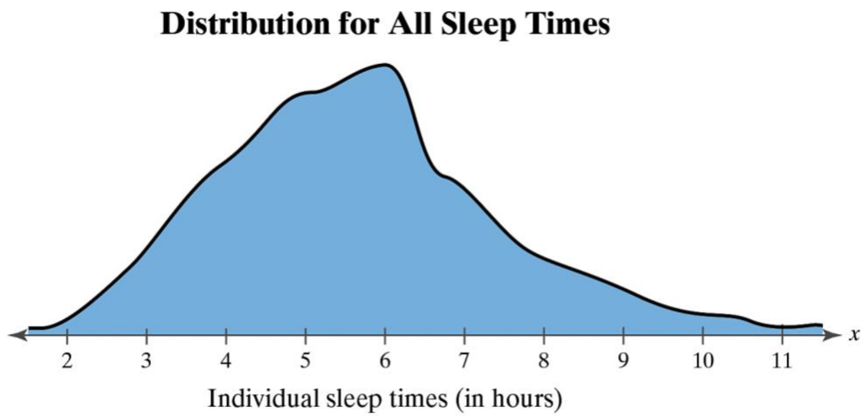
- The sampling distribution of sample means has a variance equal to $1/n$ times the variance of the population and a standard deviation equal to the population standard deviation divided by the square root of n.

$$\sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \qquad \text{Variance of the sample means}$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \qquad \text{Standard deviation of the sample means (Standard error of the mean)}$$

Example: Interpreting the Central Limit Theorem

A study analyzed the sleep habits of college students. The study found that the mean sleep time was 6.9 hours, with a standard deviation of 1.5 hours. Random samples of 100 sleep times are drawn from this population, and the mean of each sample is determined. Find the mean and standard deviation of the sampling distribution of sample means. Then sketch a graph of the sampling distribution. *(Adapted from National Institutes of Health)*



Solution: Interpreting the Central Limit Theorem

- The mean of the sampling distribution is equal to the population mean

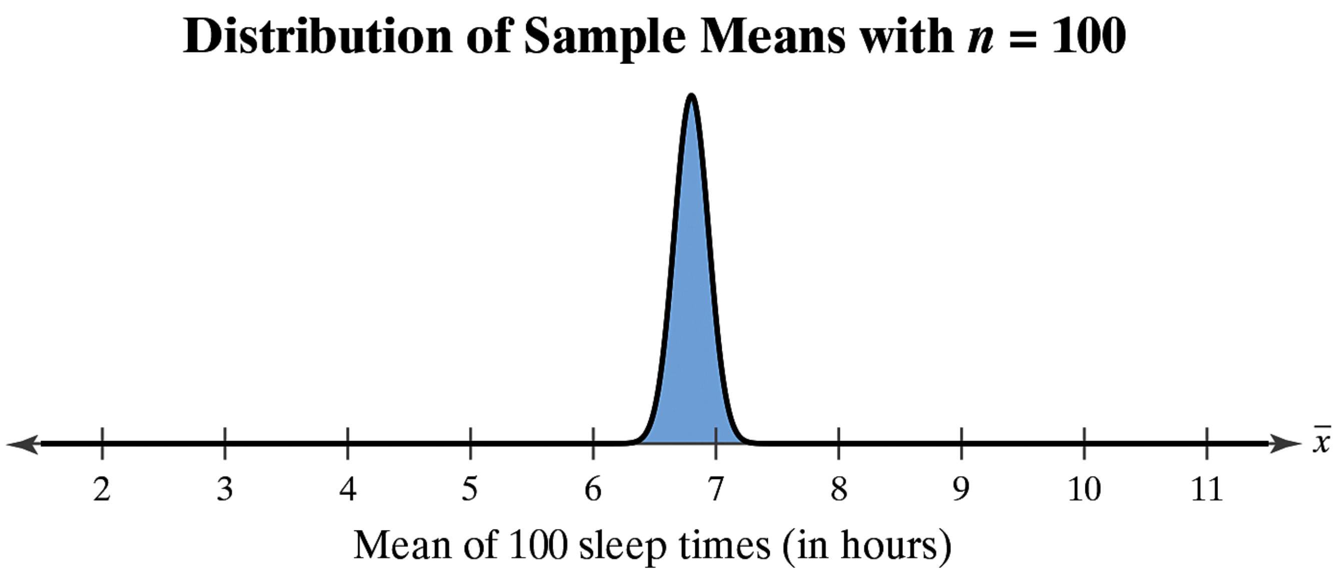
$$\mu_{\bar{x}} = \mu = 6.9$$

- The standard error of the mean is equal to the population standard deviation divided by $\sqrt{n}$.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{1.5}{\sqrt{100}} = 0.15.$$

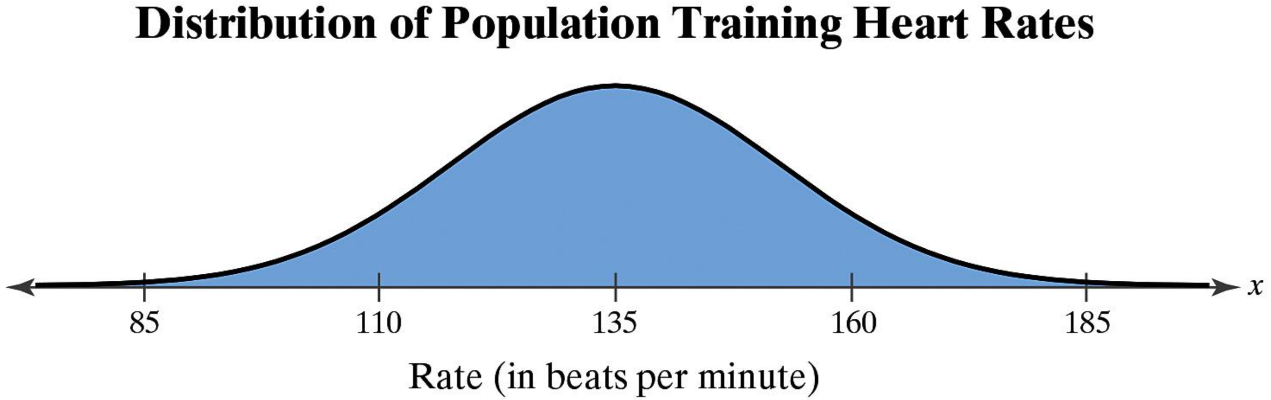
Solution: Interpreting the Central Limit Theorem

- Since the sample size is greater than 30, the sampling distribution can be approximated by a normal distribution with a mean of 6.9 hours and a standard deviation of 0.15 hour.



Example: Interpreting the Central Limit Theorem

The training heart rates of all 20-years old athletes are normally distributed, with a mean of 135 beats per minute and standard deviation of 18 beats per minute. Random samples of size 4 are drawn from this population, and the mean of each sample is determined. Find the mean and standard error of the mean of the sampling distribution. Then sketch a graph of the sampling distribution of sample means.



Solution: Interpreting the Central Limit Theorem

- The mean of the sample means

$$\mu_{\bar{x}} = \mu = 135 \text{ beats per minute}$$

- The standard deviation of the sample means

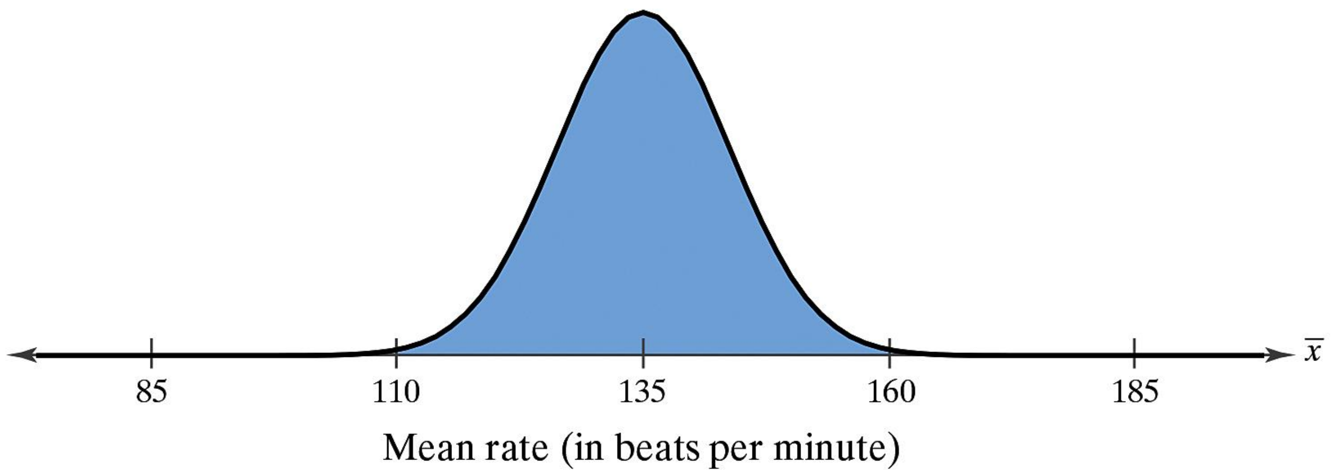
$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{18}{\sqrt{4}} = 9 \text{ beats per minute}$$

Solution: Interpreting the Central Limit Theorem

- Since the population is normally distributed, the sampling distribution of the sample means is also normally distributed.

$$\mu_{\bar{x}} = 135 \qquad \sigma_{\bar{x}} = 9$$

Distribution of Sample Means with *n* = 4



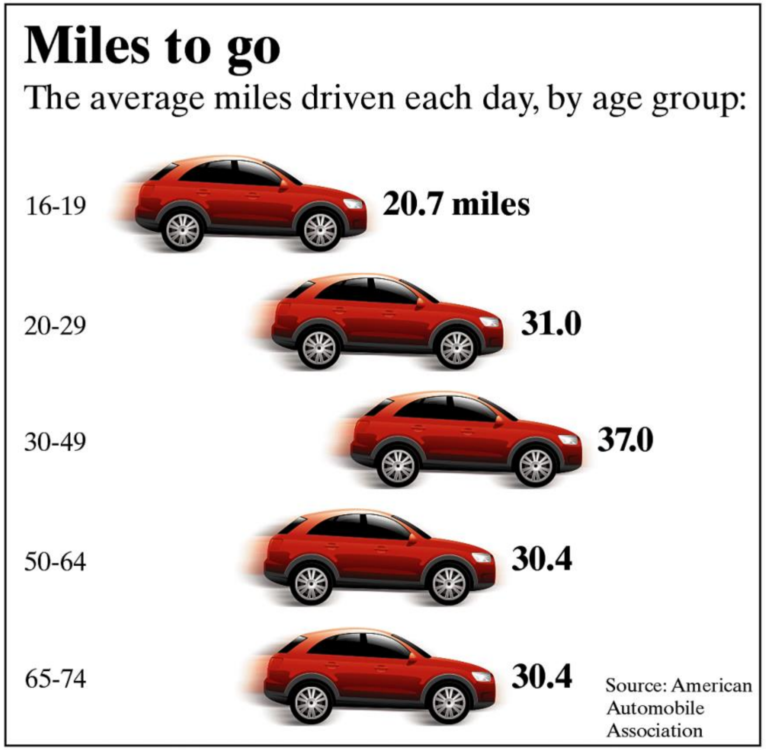
Probability and the Central Limit Theorem

- To transform $\bar{x}$ to a z-score

$$z = \frac{\text{Value} - \text{Mean}}{\text{Standard Error}} = \frac{\bar{x} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{\bar{x} - \mu}{\sigma / \sqrt{n}}$$

Example: Finding Probabilities for Sampling Distributions

The figure shows the mean distances traveled by drivers each day. You randomly select 50 drivers ages 16 to 19. What is the probability that the mean distance traveled each day is between 19.4 and 22.5 miles? Assume $\sigma = 6.5$ miles.

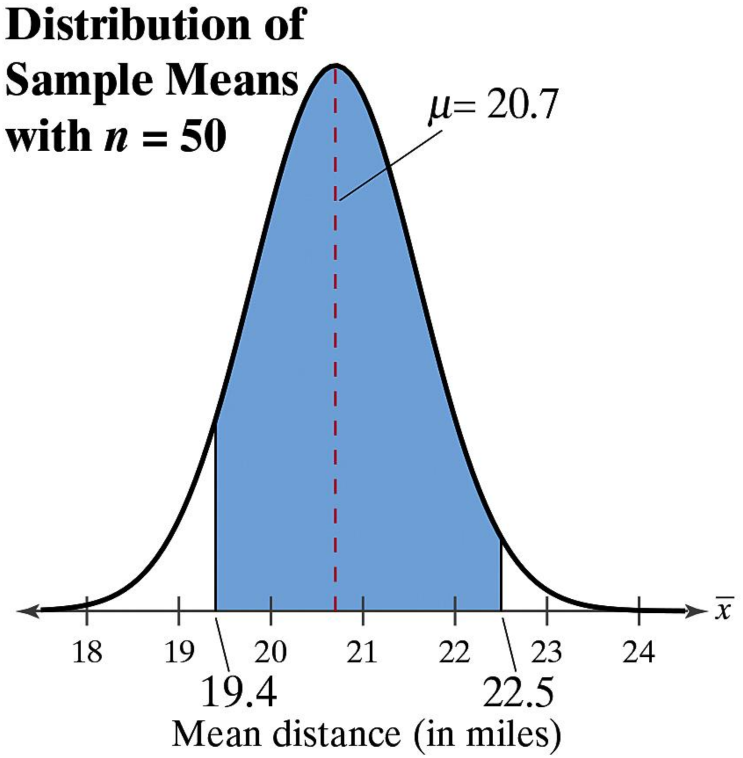


Solution: Finding Probabilities for Sampling Distributions

From the Central Limit Theorem (sample size is greater than 30), the sampling distribution of sample means is approximately normal with

$$\mu_{\bar{x}} = \mu = 20.7 \text{ miles} \quad \text{and}$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{6.5}{\sqrt{50}} \approx 0.9 \text{ miles}$$



Solution: Finding Probabilities for Sampling Distributions

- The z-scores that correspond to sample means of 19.4 and 22.5 miles are found as shown.

$$z_1 = \frac{19.4 - 20.7}{6.5/\sqrt{50}} \approx -1.41$$

$$z_2 = \frac{22.5 - 20.7}{6.5/\sqrt{50}} \approx 1.96$$

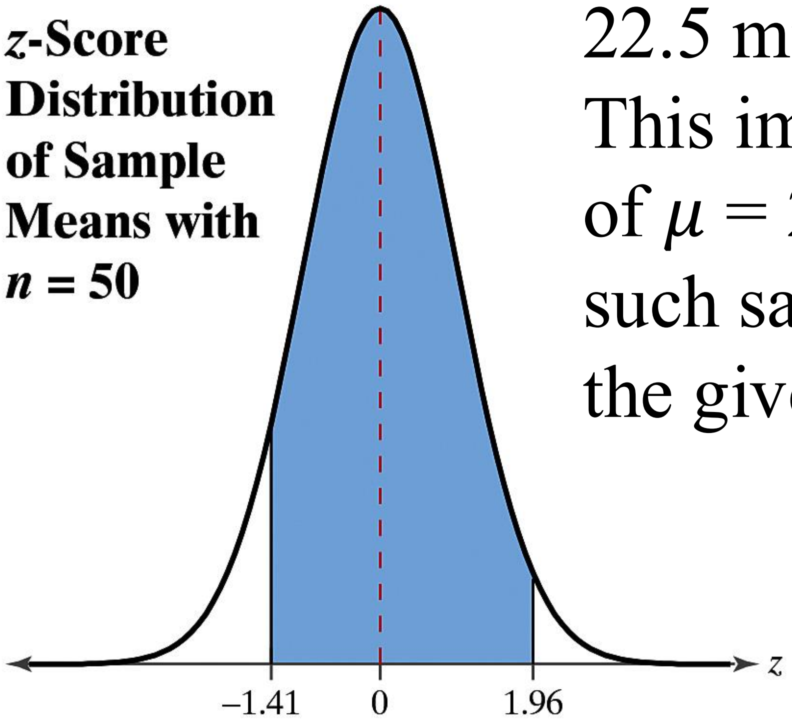
Solution: Finding Probabilities for Sampling Distributions

- The probability that the mean distance driven each day by the sample of 50 people is between 19.4 and 22.5 miles is

$$P(19.4 < \bar{x} < 22.5) = P(-1.41 < z < 1.96)$$
$$= P(z < 1.96) - P(z < -1.41)$$
$$= 0.9750 - 0.0793$$
$$= 0.8957$$

Solution: Finding Probabilities for Sampling Distributions

Of all samples of 50 drivers ages 16 to 19, about 90% will drive a mean distance each day between 19.4 and 22.5 miles, as shown in the graph. This implies that, assuming the value of $\mu = 20.7$ is correct, about 10% of such sample means will lie outside the given interval.



Example: Finding Probabilities for Sampling Distributions

The mean room and board expense per year at four-year colleges is \$11,806. You randomly select 9 four-year colleges. What is the probability that the mean room and board is less than \$12,250? Assume that the room and board expenses are normally distributed with a standard deviation of \$1650. *(Adapted from National Center for Education Statistics)*

Solution: Finding Probabilities for Sampling Distributions

- Because the population is normally distributed, you can use the Central Limit Theorem to conclude that the distribution of sample means is normally distributed, with a mean and a standard deviation of

$$\mu_{\bar{x}} = \mu = \$11{,}806 \text{ and } \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{\$1650}{\sqrt{9}} = = \$550.$$

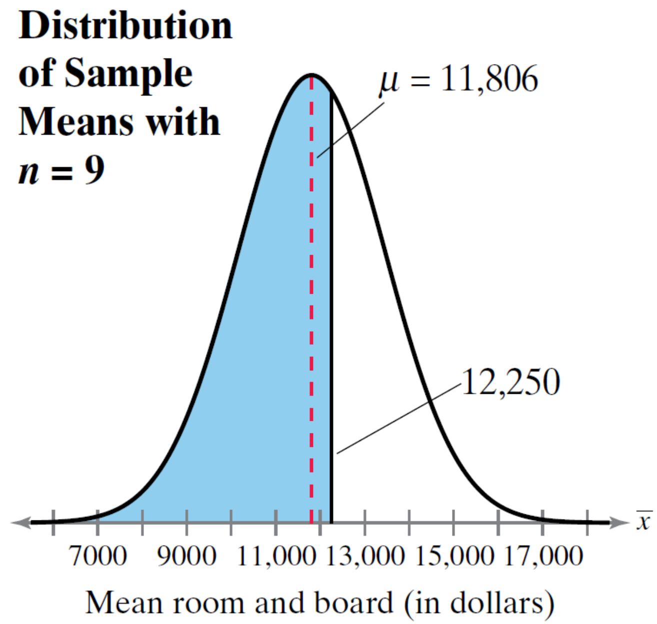
Solution: Finding Probabilities for Sampling Distributions

- The area to the left of \$12,250 is shaded. The z-score that corresponds to \$12,250 is

$$z = \frac{12{,}250 - 11{,}806}{1650/\sqrt{9}} = \frac{444}{550} = 0.81.$$

- So, the probability that the mean room and board expense is less than \$12,250 is

$$P(\bar{x} < 12{,}250) = P(z < 0.81) = 0.7910.$$



Solution: Finding Probabilities for Sampling Distributions

You can check this answer using technology. For instance, you can use a TI-84 Plus to find the x-value,

TI-84 PLUS

```
normalcdf(-10000,12250,
11806,550)
                0.7902453767
```

So, about 79% of such samples with $n = 9$ will have a mean less than \$12,250 and about 29% of these sample means will be greater than \$10,750.

Example: Finding Probabilities for *x* and $\bar{x}$

Some college students use credit cards to pay for school-related expenses. For this population, the amount paid is normally distributed, with a mean of \$2172 and a standard deviation of \$740. *(Adapted from Sallie Mae/Ipsos)*

1. What is the probability that a randomly selected college student, who uses a credit card to pay for school-related expenses, paid less than \$1900?
2. You randomly select 25 college students who use credit cards to pay for school-related expenses. What is the probability that their mean amount paid is less than \$1900?
3. Compare the probabilities from parts 1 and 2.

Solution: Finding Probabilities for x and $\bar{x}$

1. In this case, you are asked to find the probability associated with a certain value of the random variable x. The z-score that corresponds to x = \$1900 is

$$z = \frac{x - \mu}{\sigma} = \frac{1900 - 2172}{740} = \frac{-272}{740} \approx -0.37.$$

So, the probability that the student paid less than \$1900 is

$$P(x < 1900) = P(z < -0.37) = 0.3557.$$

Example: Finding Probabilities for *x* and $\bar{x}$

2. Here, you are asked to find the probability associated with a sample mean *x*. The *z*-score that corresponds to *x* = $1900 is

$$z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}} = \frac{1900 - 2172}{740/\sqrt{25}} = \frac{-272}{148} = -1.84$$

So, the probability that the mean credit card balance of the 25 card holders is less than $1900 is

$$P(\bar{x} < 1900) = P(z < -1.84) = 0.0329.$$

Example: Finding Probabilities for x and $\bar{x}$

You can check the answers for part 1 and 2 using technology. For instance, you can use Excel to find the probabilities.

EXCEL

	A
1	0.356597851

=NORM.DIST(1900,2172,740,TRUE)

EXCEL

	A
1	0.033043152

=NORM.DIST(1900,2172,148,TRUE)

Example: Finding Probabilities for x and $\bar{x}$

3. Although there is about a 36% chance that a college student who uses a credit card to pay for school-related expenses will pay less than \$1900, there is only about a 3% chance that the mean amount a sample of 25 college students will pay is less than \$1900. Because there is only a 3% chance that the mean amount a sample of 25 college students will pay is less than \$1900, this is an unusual event.